John Poirier

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EDUCATION

Northeastern University, Boston, MA — September 2026 - Expected May 2028
Master of Science in Robotics - Concentration in Electrical Engineering
Relevant Courses: Robot Mechanics and Control, Mobile Robotics

University of Vermont, Burlington, VT — August 2022 - May 2026
Bachelor of Science in Electrical Engineering, Certificate in Autonomy and Robotics, Summa Cum Laude
Relevant Courses: Power Electronics, Microcontroller Systems, Electronics I, Signals & Systems, Controls Engineering

SKILLS

Tools & Software: Altium, LTspice, Git, Linux, AutoCAD, Revit, SolidWorks
Programming: Python, C++ (Arduino), MATLAB, Julia
Hardware & Lab: Oscilloscope, DMM, Power Supply, Soldering & Reflow, RGB+Depth Camera, NVIDIA Jetson

PROJECTS

Autonomous Weeder Robot, Capstone Project at University of Vermont — August 2025 - June 2026
- Led software and sensor integration for a capstone team developing a dual-camera ROS 2 node in Python and OpenCV to integrate RGB+Depth camera inputs on an NVIDIA Jetson
- Deployed a custom YOLO image classification model using PyTorch for real-time weed detection to drive hardware actuation, achieving 93% classification accuracy and 96% classification precision during testing
- Implemented autonomous point-to-point and path-following navigation algorithms for a differential drive robot

Line-Following Robot, Auckland University of Technology — Spring 2025
- Designed two PCBs end-to-end in Altium; a main control board and an automatic race-starter board, from schematic capture and layout through hand-soldering, reflow, and testing
- Characterized power rail performance using a test circuit and variable DC load, showing a -18dB noise reduction through a voltage regulator to ensure stabilized tone generation on race-starter
- Programmed a state-based motor controller in C++ on an Arduino Nano using IR sensor and limit switch feedback, reducing track completion time by nearly 3x across design iterations

Wireless Charging Module, Auckland University of Technology — Spring 2025
- Designed a modular resonant inductive power transfer board and a receiver test board using hand-wound coils
- Selected switching MOSFET by characterizing rise/fall times with an Ocsilloscope to ensure proper operation at the maximum 80 kHz input frequency
- Maximized power transfer efficiency at the calculated 91.15 kHz resonant frequency by varying board spacings from 2-10 mm to reach a peak efficiency of 63.9% at a 4.5 mm gap

EXPERIENCE

Electrical Engineer Intern, DuBois & King, Waterbury, VT — July 2025 - August 2026
- Drafted power, lighting, and fire alarm system layouts and power service diagrams for 20+ commercial buildings in AutoCAD and Revit
- Verified designs for National Electrical Code compliance and coordinated with mechanical and plumbing teams to resolve cross-discipline conflicts

Undergraduate Researcher, University of Vermont, Burlington, VT — Summer 2024
- Optimized renewable energy dispatch for large-scale power grid models in Julia, decreasing processing time by up to 40% and memory allocation by up to 99% as measured with BenchmarkTools
- Parallelized solver workloads using Gurobi with CUDA for multi-threaded and GPU-accelerated processing